Formlizing Quantitative Equational Reasoning

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Program Verification Exam
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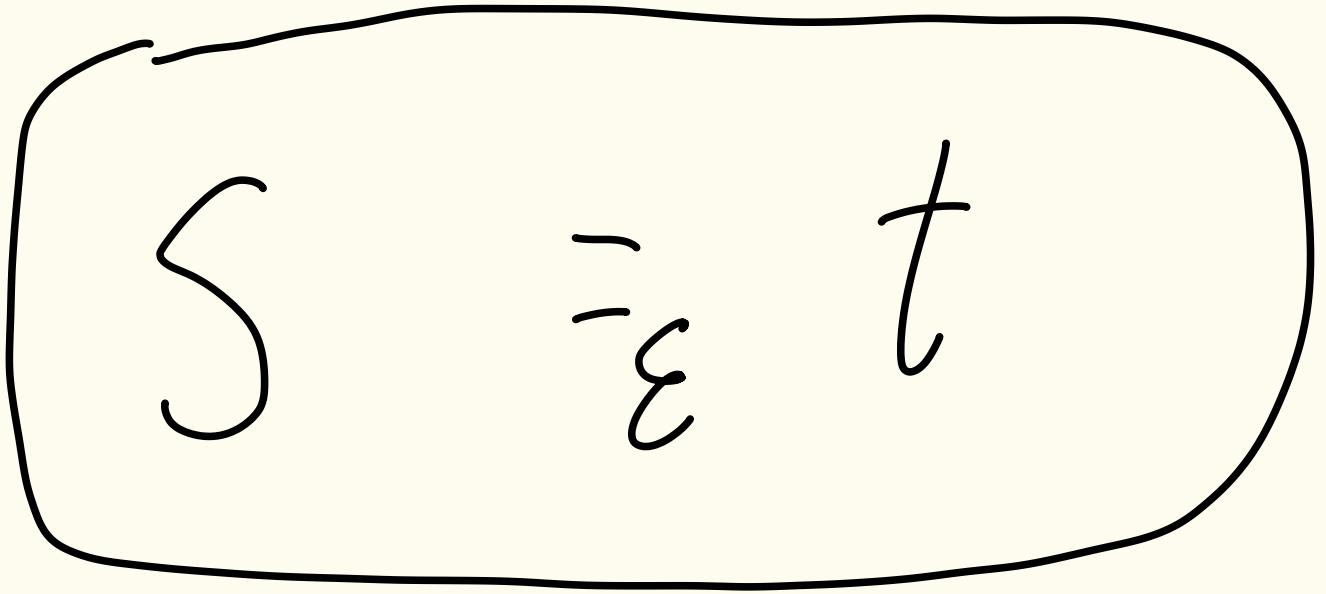
Motivation

- Equations are fundamental to mathematical reasoning

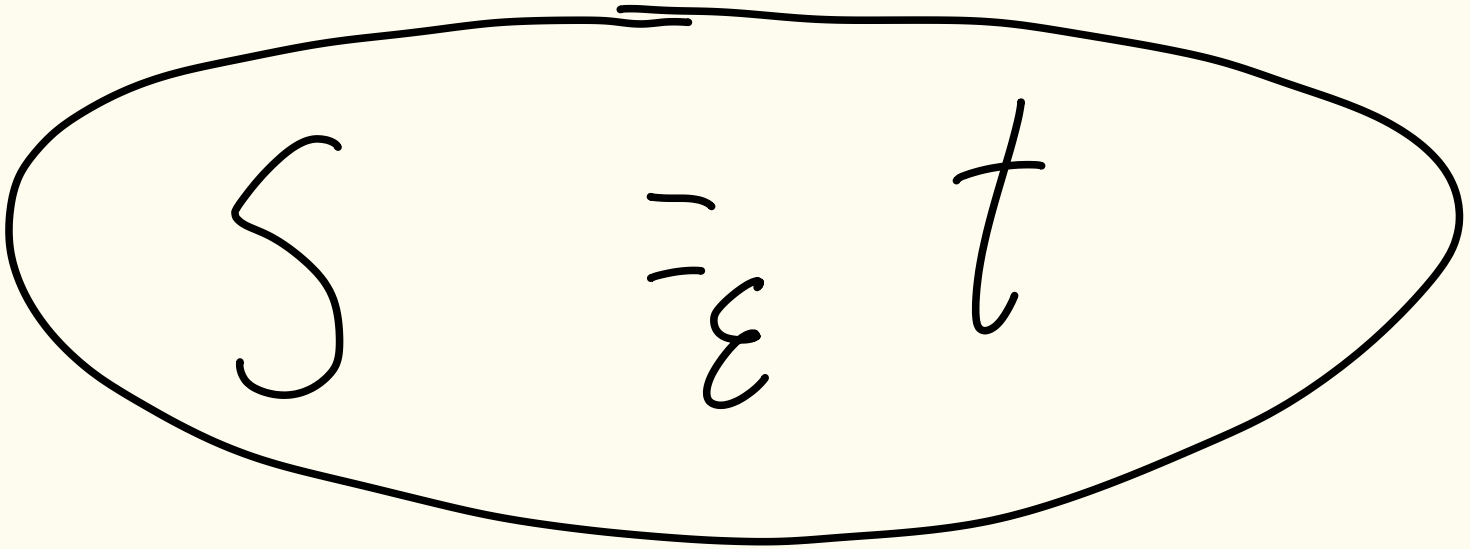
- Reasoning about programs is also based on program equivalences

- We want quantitative analogues of equational reasoning

Replace standard equations with quantitative ones



Replace standard equations with quantitative ones



"S is within ε distance of t"

An Equational Theory of Barycentric Algebras

Variables
$s, t, u \ldots$

Operations
$\{ +_p \mid 0 < p < 1 \}$

Equations

(Idempotency) $s +_p s = s$

(Skew commutativity) $s +_p t = t +_{1-p} s$

(Skew associativity) $(s +_p t) +_q u =$

$$s +_{pq} \left(t +_{\frac{(1-p)q}{1-pq}} u \right)$$

A Model of the Theory

The models interpret $\vdash_p$ while respecting axioms

One model is the finitely supported probability distributions and convex combinations

<Small Example>

How to extend to the quantitative setting?

Quantitative equational theories

operations + quantitative equations

EMet - Lawvere theories

Lawvere categories enriched in the category of extended metric spaces

Monads on Set (EMet)

Quantitative effects

Madare et al

What is a model in the quantitative setting?

If $s =_\varepsilon t$

Then for an interpretation $[\![\cdot]\!] : T_{var} \to \mathcal{M}$

$$d([\![s]\!], [\![t]\!]) \leq \varepsilon$$

Where d is a metric.

Note $s =_0 t$ means strict equality as d is a metric.

A generalization of this relaxes the metric requirement to simply fuzzy relations

$$d : A \times A \to [0,1]$$

No additional assumptions

But then we still need $=$ to denote exact equality as $s =_0 t$ may not imply $[\![s]\!] = [\![t]\!]$.

Satisfiability Relation

Let $\mathcal{A}$ be a quantitative algebra

$\mathcal{A} \models \forall (B, d_B).\ s = t$ iff $[\![s]\!] = [\![t]\!]$

$\mathcal{A} \models \forall (B, d_B).\ s =_\varepsilon t$ iff $d_{\mathcal{A}}([\![s]\!], [\![t]\!]) \leq \varepsilon$

For all 1-Lipschitz $[\![\cdot]\!] : B \to \mathcal{A}$

A quantitative theory of Barycentric algebras

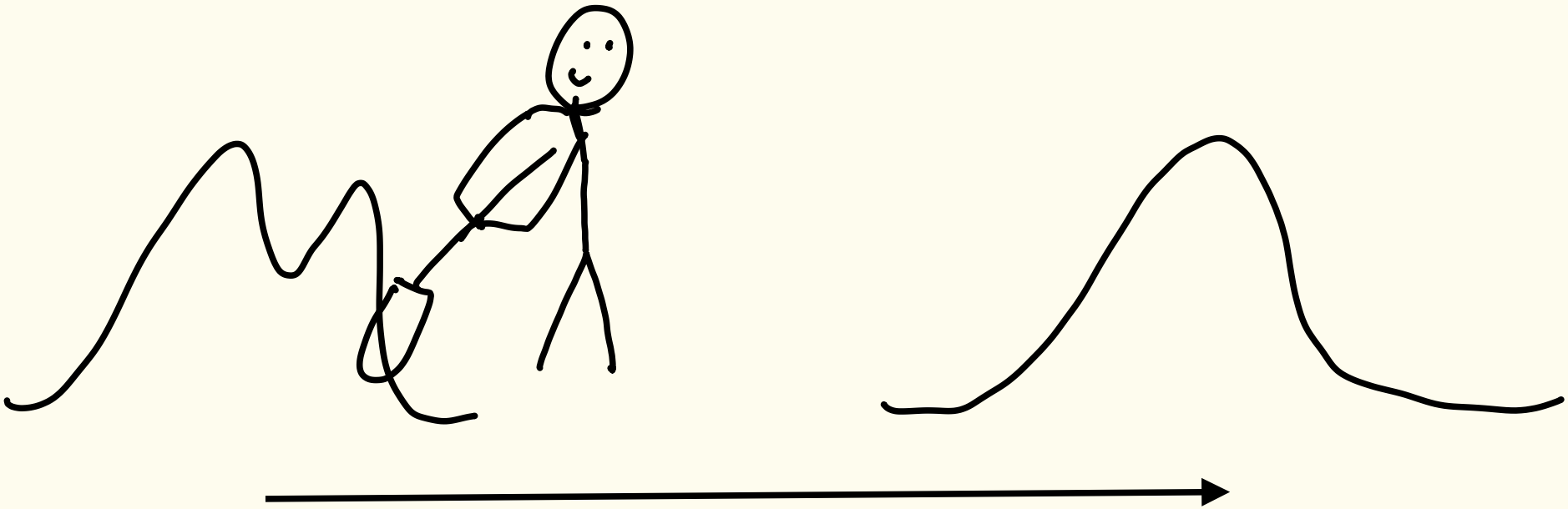
(Idempotency) $x[\overset{x}{1}] \cdot x +_p x = x$

(Skew comm) $\begin{matrix} & x & q \\ x & 1 & 1 \\ y & 1 & 1 \end{matrix} \cdot \; x +_p y = y +_{1-p} x$

(Skew assoc) $\begin{matrix} & x & y & z \\ x & 1 & 1 & 1 \\ y & 1 & 1 & 1 \\ z & 1 & 1 & 1 \end{matrix} \cdot (x +_p y) +_q z = x +_{pq} (y +_{\frac{(1-p)q}{1-pq}} z)$

(Interpolative) $\begin{matrix} & x & y & w & z \\ x & 1 & 1 & \varepsilon & 1 \\ y & 1 & 1 & 1 & \delta \\ w & 1 & 1 & 1 & 1 \\ z & 1 & 1 & 1 & 1 \end{matrix} \cdot x +_p y = {}^{p\varepsilon + (1-p)\delta} w +_p z$

An aside about moving piles of sand



Out of all the possible ways to move the sand, which is the least amount of work?

$$K(\mu,\nu) = \inf_{\pi \in C(\mu,\nu)} \int d(x,y)\, d\pi$$

optimal transport

A model in the quantitative setting

Finitely supported probability distributions, their convex sums and the Kantorovich "lifting" is a model.

Actually it is Axiomatised

If $s =_\varepsilon t$ is derivable then $K([\![s]\!], [\![t]\!]) \leq \varepsilon$

If $K([\![s]\!], [\![t]\!]) \leq \varepsilon$ then $s =_\varepsilon t$ is derivable

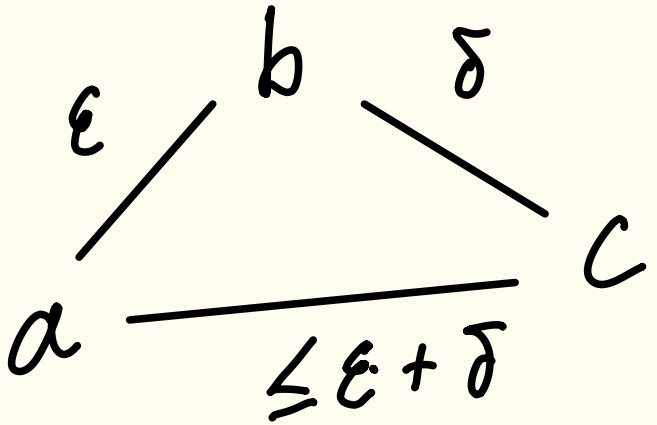
General axioms

Some things should always hold

$$\frac{a = b \qquad b = c}{a = c}$$

Does not quite hold for $=_\varepsilon$

$$\frac{a =_\varepsilon b \qquad b =_\delta c}{a =_{\varepsilon+\delta} c}$$



... This is only general in the metric view.

Other general axioms

$$\frac{\varepsilon \leq \delta \quad a =_\varepsilon b}{a =_\delta b} \qquad \frac{}{a =_1 b}$$

$$\frac{\{ s =_\delta t \mid \varepsilon < \delta \}}{s =_\varepsilon t}$$

Proven to be not needed for Barycentric algebras

A model

Let (A, d) be a fuzzy relation and $(\mathcal{D}(A), \{+_p \mid 0 < p < 1\})$ be the convex algebra of finitely supported probability distributions.

Then the quantitative algebra

$$(\mathcal{D}(A), \{+_p \mid 0 < p < 1\}, q_{K_d})$$

is a model

Soundness

Let $(\mathcal{A}, d)$ be a fuzzy relation and s, t be two Barycentric algebra terms with variables in $\mathcal{A}$ then

If $\mathcal{LC}_{\mathcal{A}} \vdash \forall (x, y \in \mathcal{A}).\ s =_\varepsilon t$ is derivable

Then $K_{d_d}(\llbracket s \rrbracket, \llbracket t \rrbracket) \leq \varepsilon$

< Note to self need to pin down difference between $\vdash$ and $\models$ >

Completeness

Let $(\mathcal{A}, d)$ be a fuzzy relation and s, t be two Bargcentric algebra terms with variables in $\mathcal{A}$ then

If $K_{d_{\mathcal{A}}}([\![s]\!], [\![t]\!]) \leq \varepsilon$

Then $\mathcal{LC}_{\mathcal{A}} \vdash \forall(\mathcal{A}, d_{\mathcal{A}}).\ s \approx_{\varepsilon} t$ is derivable

```
Record signature := {
  sym : Type;
  arity : sym -> nat;
}.

Inductive term (sig : signature) (X : Type) : Type :=
  | Var : X -> term sig X
  | App : forall f : sym sig, ('I_(arity f) -> term sig X) -> term sig X.
```

```
Record fuzzy_space (R : realType) := {
    fcarrier : Type;
    frel : fcarrier -> fcarrier -> R;
    frel_range :
      forall x y, (0 <= frel x y <= 1)%R;
}.

Record FuzzyAlgebra (R : realType) (sig : signature) := {
  fa_space : fuzzy_space R;
  fa_ops : algebra_ops sig (fcarrier fa_space)
}.
```

```
Inductive judgement {R : realType} (sig :
signature) (X : Type) :=
  | EqJ : term sig X -> term sig X ->
judgement sig X
  | QEqJ : R -> term sig X -> term sig X ->
judgement sig X.
```

```
Definition satisfies {R : realType} {sig}
    (A : FuzzyAlgebra R sig)
    (X : fuzzy_space R)
    (phi : judgement sig (fcarrier X)) : Prop :=
  forall rho : interpretation X A,
    match phi with
    | EqJ s t => fuzzy_eval A rho s = fuzzy_eval A rho t
    | QEqJ eps s t =>
      (frel (fa_space A)
        (fuzzy_eval A rho s)
        (fuzzy_eval A rho t) <= eps)%R
    end.

Definition fuzzy_theory (R : realType) (sig : signature) :=
  forall X : fuzzy_space R,
    @judgement R sig (fcarrier X) -> Prop.

Definition fuzzy_models {R : realType} {sig}
    (A : FuzzyAlgebra R sig)
    (E : fuzzy_theory R sig) : Prop :=
  forall X phi, E X phi -> satisfies A X phi.
```

<Frel deduction system only
show a couple of the strict
equality rules>

<Lemma frel_derives weakening>

<Theorem derives_fin_sound>

<ICA signature and related>

<Inductive ica_theory>

<Important stuff from probability distribution file>

<Important parts of soundness file>

<Current state of completeness and how to proceed>

How does this relate to quantum?

- In my PhD i am trying to apply this line of thinking to quantum computing

- We want to start small by finding a theory for mixed quantum states

- Later we would like a theory for programs

Why mixed states?

- They can be viewed as probability distributions over pure states

- They are a model of non-quantitative Bargcentric axioms.

- There exists a notion of quantum optimal transport

$$T_c(\rho,\sigma) = \inf_{\Gamma \in C(\rho,\sigma)} \operatorname{tr}(C\Gamma)$$

Should be easy, right?

- Turns out that Barycentric structure can only reveal the transportation of mass.
- Mixed states can be coupled in many different ways that transports the mass the same, but $tr(CI)$ gives different values
- Currently using a mix of graphs and linear algebra which seems promising.

<Conclusion>